

Too many knots

Abstract

Introduction

Recent computational advances have allowed for the rapid estimation of large spatial and spatiotemporal models in ecology and related fields (Banerjee & Fuentes 2012). Motivating questions for these applications to large georeferenced datasets including tracking abundance in time and space, identifying environmental predictors of occurrence, or quantifying range shifts in distribution (Ross *et al.* 2012; Miller *et al.* 2013; Thorson & Barnett 2017).

There are a number of statistical approaches for developing models with spatial components; examples include generalized additive models (GAMs) (Murase *et al.* 2009; Forney *et al.* 2012), non-parametric approaches (Mutanga *et al.* 2012), and applications of Gaussian random fields (Latimer *et al.* 2009; Datta *et al.* 2016). Advances by Rue *et al.* (2009) have enabled the development of the Integrated Nested Laplace Approximation (INLA) as a fast and efficient approximation to estimating high-dimensional Gaussian random fields. These models have been extended to include the ability to fit geographic barriers, such as islands or lakes, anisotropic covariance functions, and non-Gaussian data (Bakka *et al.* 2018). In addition to estimating Bayesian posterior distributions (Lindgren & Rue 2015), INLA has been integrated with Template Model Builder (TMB; (Kristensen *et al.* 2016; Osgood-Zimmerman & Wakefield 2021)) to enable auto-differentiation and has been adopted in many spatial fisheries applications around the world (Thorson 2019b; e.g., Johnson *et al.* 2019).

Constructing spatiotemporal models with INLA or related software involves many of the same considerations as with simpler non-spatial models. For example, modelers frequently ask: “Which covariates are most important?” and “Does my model do a satisfactory job of making predictions?” Spatial or spatiotemporal models may include additional considerations, such as specifying the form of spatiotemporal variation (Anderson & Ward 2019). One of the more complicated and important decisions—particularly for INLA based models—lies in specifying the dimensionality of the latent spatial fields (Lindgren & Rue 2015). Unlike Generalized Additive Models (GAMs), which often penalize smooth terms (Wood 2003), penalties in overfitting

INLA models are not as apparent.

One of the first steps in fitting a model with INLA involves choosing the appropriate resolution of the spatial field using a “mesh” (a map consisting of small triangles, e.g., Figure 1). By estimating values of the spatial field at the triangle vertices (or “knots”), the spatial field may be triangulated to coordinates where data are observed or predictions are made (Lindgren *et al.* 2011). There are a number of decision points to be made in constructing the mesh, including whether to include a boundary, whether the boundary is regular, the maximum edge of any triangle (in the core or boundary area), and the minimum angles allowed by the triangles (Gómez-Rubio). Each of these decisions can impact the density and number of small triangles that make up the mesh—and in turn, the accuracy of predictions. Though the mesh construction is important, this step has been omitted from best practices guides using an INLA-based approach (Thorson 2019a), and we are only aware of one sensitivity analysis examining the importance of mesh construction (Righetto *et al.* 2020). While previous research has suggested that increasing the resolution of the mesh improves predictive accuracy (Thorson 2019a; Godana *et al.* 2019; Wilson 2021; Williamson *et al.*), other authors have also noted that finer meshes don’t always translate into better performance (Huang *et al.* 2017).

The objective of this paper is to demonstrate that higher resolution spatial approximations in INLA may not translate to better predictive accuracy. Using a series of analyses on a real-world case study of groundfish data in the Northeast Pacific Ocean, we evaluate the out-of-sample predictive performance as a function of mesh resolution (number of vertices). We further evaluate these results across a range of cross-validation sampling schemes, including an extreme case of leave-one-out cross validation (LOOCV). All data for our analysis are publicly available, and code and data to reproduce our analysis is available on Github (XXXX) and Zenodo (on publication).

Methods

Data

As a case study to evaluate the effect of mesh resolution on predictive ability of INLA-based models, we use a publicly available dataset of catch rates from scientific surveys in the Northeast Pacific Ocean (<https://www.nwfsc.noaa.gov/data>). The West Coast Bottom Trawl Survey (WCBTS) dataset represents a fishery-independent survey collected annually over a 16-year period, 2003-2018, covering the California Current ecosystem on the west coast of the USA (Keller *et al.* 2017). The survey is designed to estimate the abundance, sizes, and age composition of commercially and recreationally targeted groundfish species found in near-bottom habitats. The fundamental methods of the survey, sampling intensity, gear, seasonal and

spatial coverage, and random stratified design have remained relatively constant. Because of the large number of model runs needed, we restricted our analysis to only use data from 2018 ($n = 702$ observations). Though hundreds of species we encountered in the survey, we also restrict our analysis to only include data from four species that are of interest to fisheries, but also represent a range of occurrences: Dover Sole (*Microstomus pacificus*, encountered in 593 tows), Petrale Sole (*Eopsetta jordani*, 291 tows), Lingcod (*Ophiodon elongatus*, 225 tows), and Darkblotched Rockfish (*Sebastes crameri*, 105 tows).

Modeling

Because of the presence of zeros and positive continuous values in fish survey datasets such as the WCBTS data, conventional approaches to analyzing these data have used a delta or hurdle model to separately model occurrence and positive catches (Thorson 2019a; e.g., Johnson *et al.* 2019). We adopt a similar approach here, using a binomial distribution with a logit link to model species occurrences, and a Gamma distribution with a log link to model positive catch rates. For simplicity and consistency with fisheries assessments using these data, we fit models estimating an intercept and spatial field throughout, e.g.

$$g(u_s) = B_0 + \omega_s$$

where ω_s is the estimated spatial field, B_0 is an intercept, u_s is the expected value for point in space s , and $g()$ is an inverse link function.

All of the meshes in our analysis used a non-convex boundary and offset based on 5% of the data (Figure 1). We fixed the maximum edge of triangles in the mesh at 100km in the core area and 500km closer to the boundary (Figure 1). Throughout our analyses, we adjusted the mesh resolution by varying the cutoff distance (representing the distance where two points are treated as a single vertex) because the cutoff has been shown to have the greatest impact on measures of model performance from mesh construction (Righetto *et al.* 2020). We fit all models using R-INLA (Lindgren & Rue 2015; R Core Team 2021) and R 4.1.0 (R Core Team 2021) and specifically the ‘inlabru’ R package, which is a convenient wrapper for the INLA software (Bachl *et al.* 2019). We used penalized complexity priors on the Matérn covariance parameters (Simpson *et al.* 2015; Osgood-Zimmerman & Wakefield 2021) with TODO (AND OTHER PRIORS).

Sensitivities

As a preliminary evaluation of whether the spatial resolution of INLA-based models can result in overfitting, we evaluated 40 cutoff values for each species and hurdle sub-model included in our analysis (cutoff values ranging from 3–120km). There is an inverse relationship between cutoff values and mesh resolution or vertices; while the results varied slightly by species, our meshes ranged in size from 75–1348 vertices. The high-resolution meshes with more vertices than observations were included as an upper extreme, though many applications of INLA to similar survey data have used 500–600 vertices (e.g., Thorson 2019a; Tolimieri *et al.* 2020).

To evaluate model performance, we conducted 10-fold cross-validation. Though not a 1-dimensional dataset, the strong latitudinal gradient has driven spatial stratification in previous analyses (Keller *et al.* 2017) (Figure 1). We a priori divided the survey domain into 10 spatial blocks by latitude quantiles with approximately the same number of points in each box. Each model was run using *inlabru* (@ Bachl *et al.* 2019), and used to predict the data for the held out test data. The predictive density was calculated using the binomial or Gamma densities, and total predictive densities were calculated by summing across folds.

As a second sensitivity, we evaluated whether the particular cross-validation method used may be influencing the results. Recent work has demonstrated that for spatially structured data, creating random folds that ignore spatial autocorrelation may bias results (Valavi *et al.* 2019). Using the presence-absence data from Dover Sole, we examined the cross validation performance across a range of mesh resolutions (cutoff values ranging from 15–75, or 122–594 vertices), number of spatial blocks (19–218) and number of folds used for cross validation (10, 40). Selection of spatial blocks was done using the ‘blockCV’ package in R (Valavi *et al.* 2019) with varying distance bands between test and training sets (25–150km).

To further validate the inference from the spatial cross-validation, we examined the performance of out-of-sample approximations in INLA. We focused on the conditional predictive ordinate (CPO), which is an approximation to the leave-one-out cross-validation score (LOOCV). Ideally, we would use cross validation to evaluate the out of sample predictive accuracy of a model. The total predictive log density can be represented as the sum of individual log posterior probabilities, $\sum_{i=1}^n \log[P(Y_i|Y_{\neq i})]$, where $P(Y_i|Y_{\neq i})$ is the probability of test data i . Because conducting true leave-one-out cross-validation may be prohibitive for large datasets, INLA approximates the out of sample posterior probability for each observation. Previous research has shown that the CPO approximation performs similarly to MCMC cross-validation (Held *et al.* 2010), yet it is less clear whether CPO is similar to true cross validation in comparing mesh resolution. Focusing on the presence-absence model of Dover Sole (Figure 2), we first calculated the total CPO score across a range of

cutoff values. For comparison, we then calculated the true predictive density using LOOCV across a similar range of cutoffs (fitting the INLA model 702 times for each value).

Results

Fig. 1 just demonstrates the data used in the study, and a couple mesh configurations (coarse and fine, cutoff of 80 and 20, translating to 110 and 394 vertices)

Fig. 2 shows the decline in performance as a function of large knots, across the 4 species. Similarly, Fig S1 shows this for the gamma distribution.

Fig. 3 shows the decline in performance as a function of large knots happens regardless of the type of sampling used (sensitivity to # blocks and # folds).

Fig. 4. shows the comparing between CPO (via INLA) and real LOOCV. LOOCV shows that performance declines as knots become too large, but CPO keeps increasing

Fig. 5 Illustrates downstream effects of knots on estimates of downstream parameters

- I think the spatial range and SD would be (most?) interesting here

Discussion

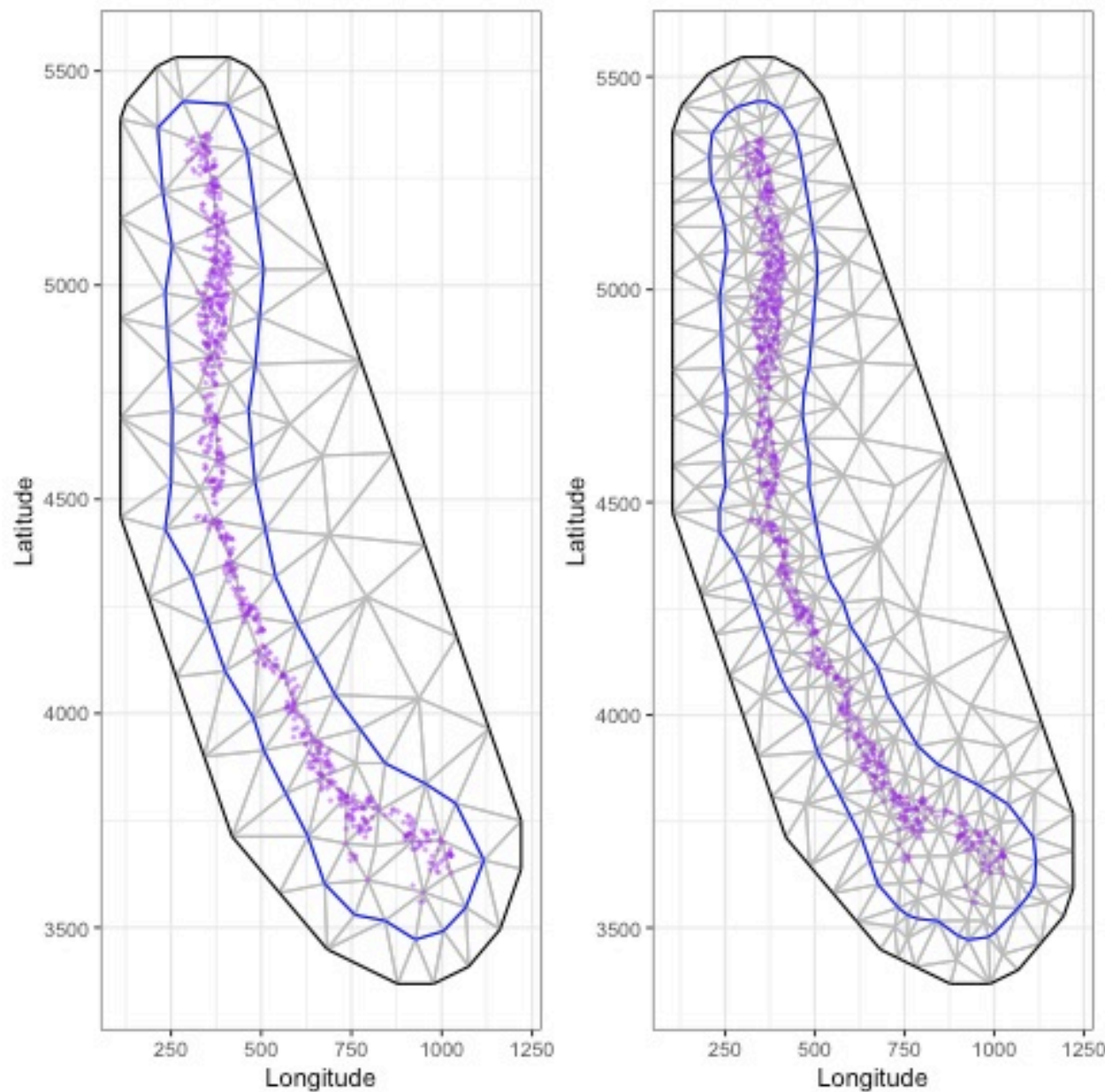
Larger meshes aren't necessarily better. Others have pointed this out, https://punama.github.io/BDI_INLA/, but it gets overlooked

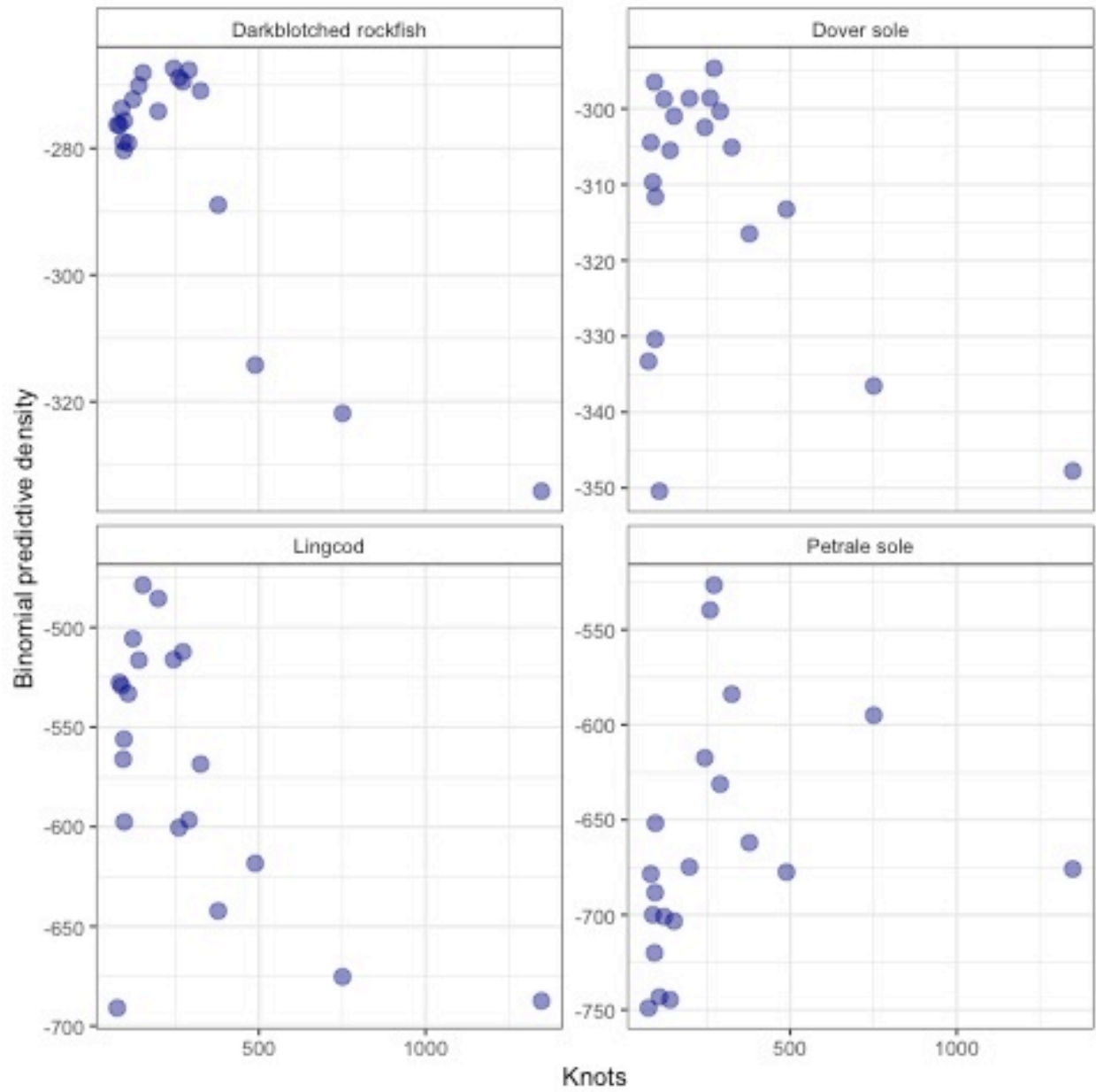
Sub-optimal meshes are likely to have an impact on derived things we care about, like indices of abundance – both bias and precision, fig s2

Datasets like the WCTBS support an optimal mesh of ~ 300 or fewer knots

We recommend analysts evaluate models across a range of resolutions and carefully consider what cross-validation scheme may be most appropriate for the question at hand

Figures

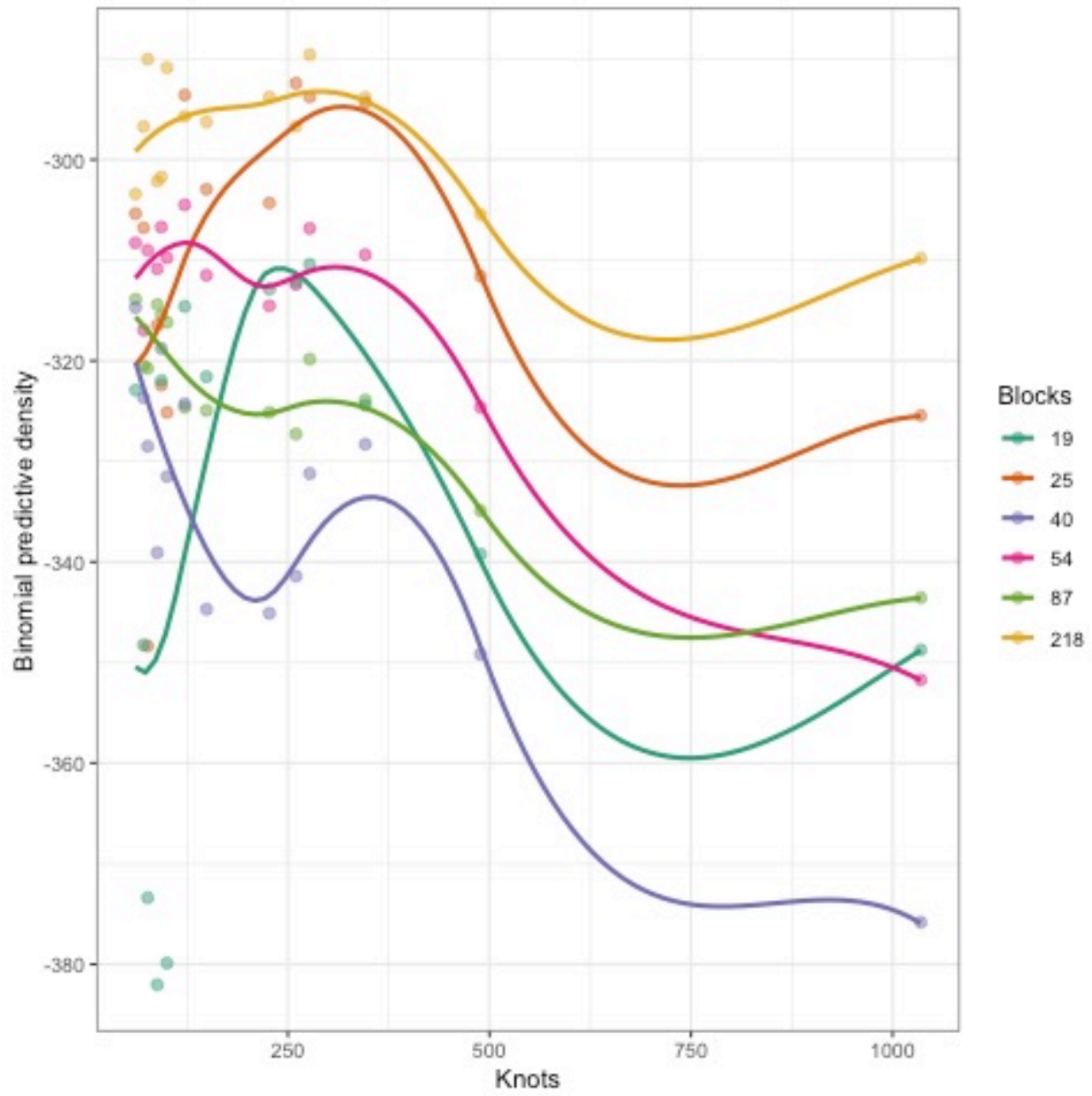




139

140

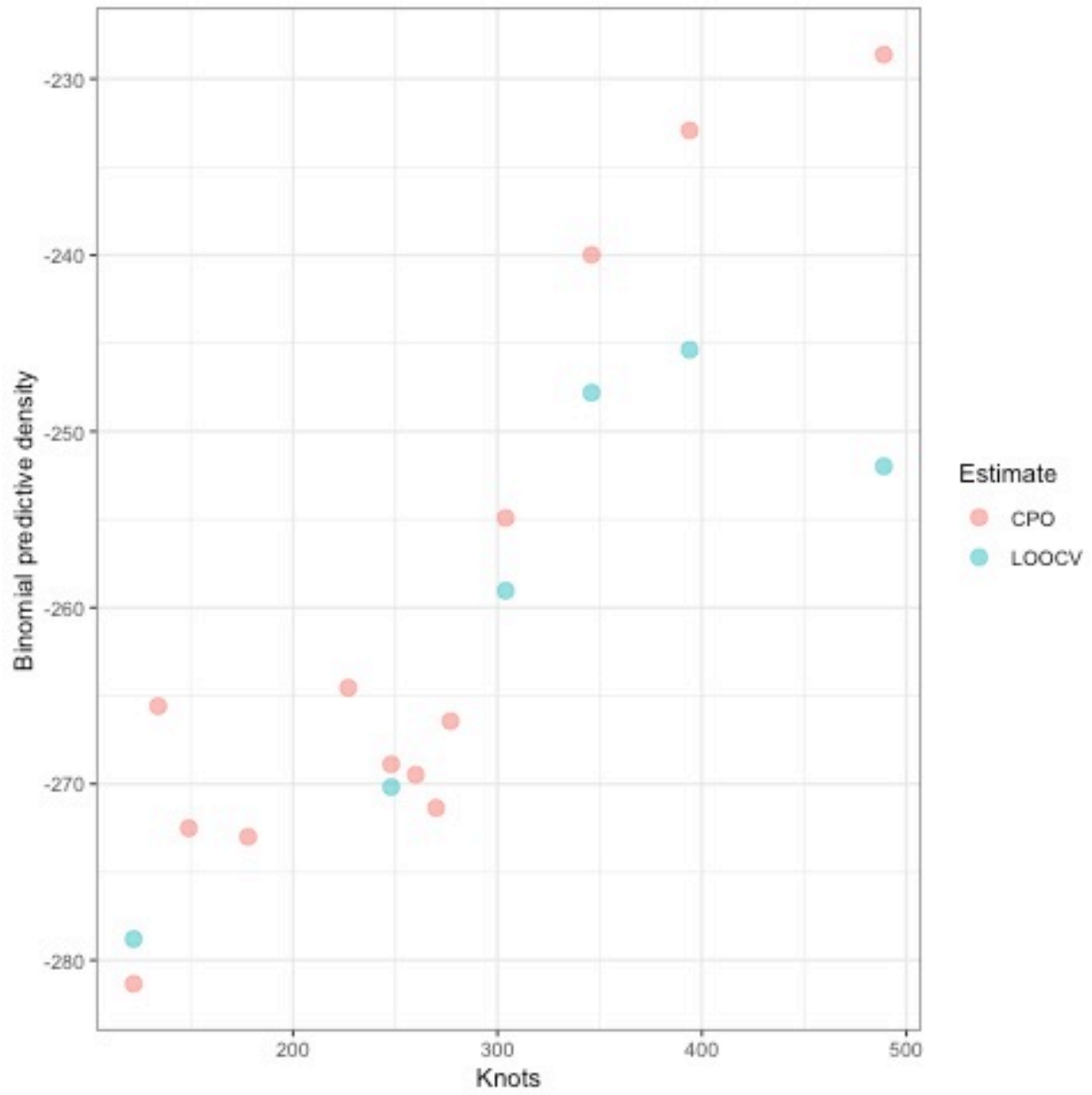
141



142

143

144



145

146

147

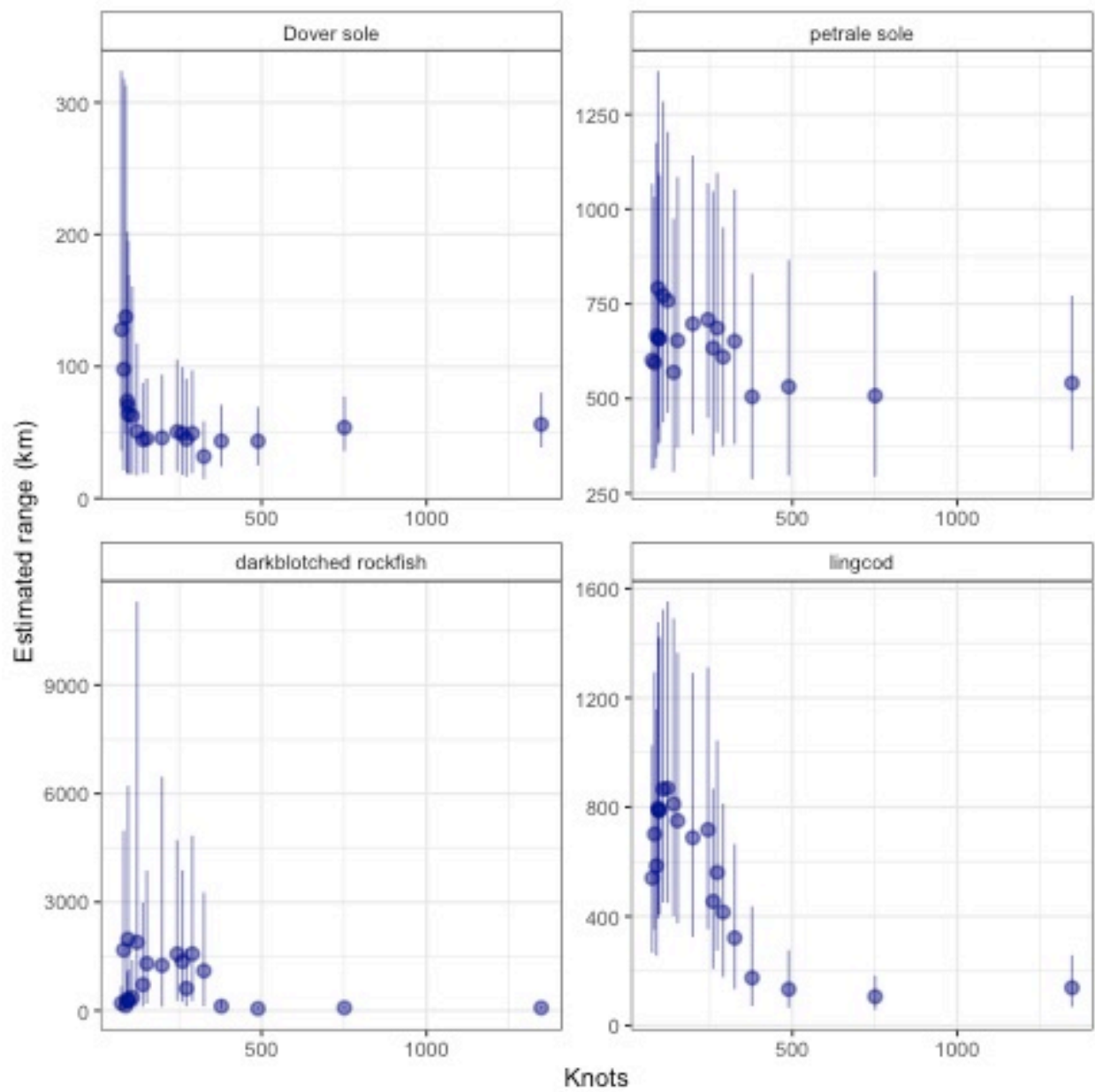


Figure 1: Figure 5

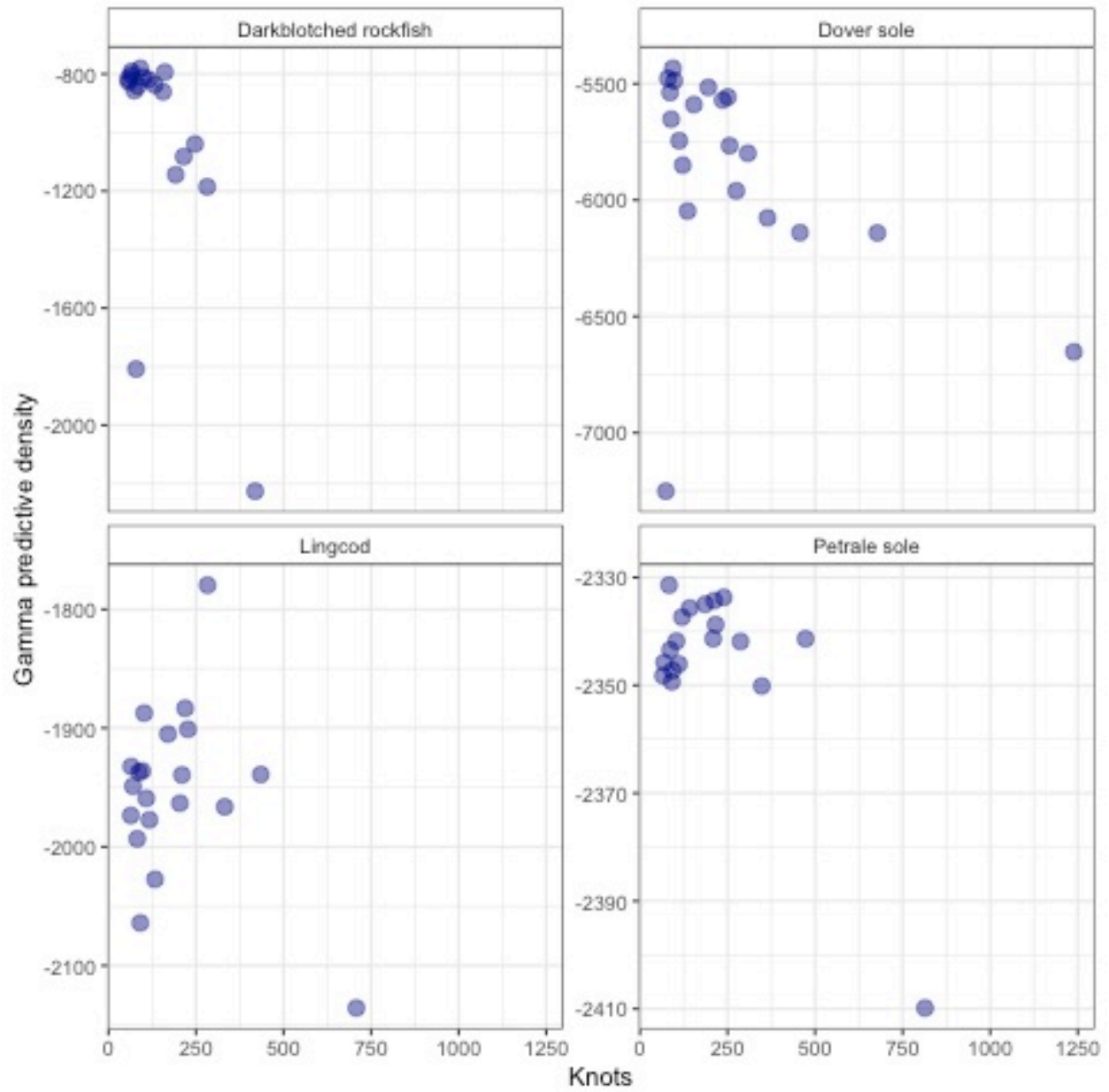


Figure 2: Figure S1

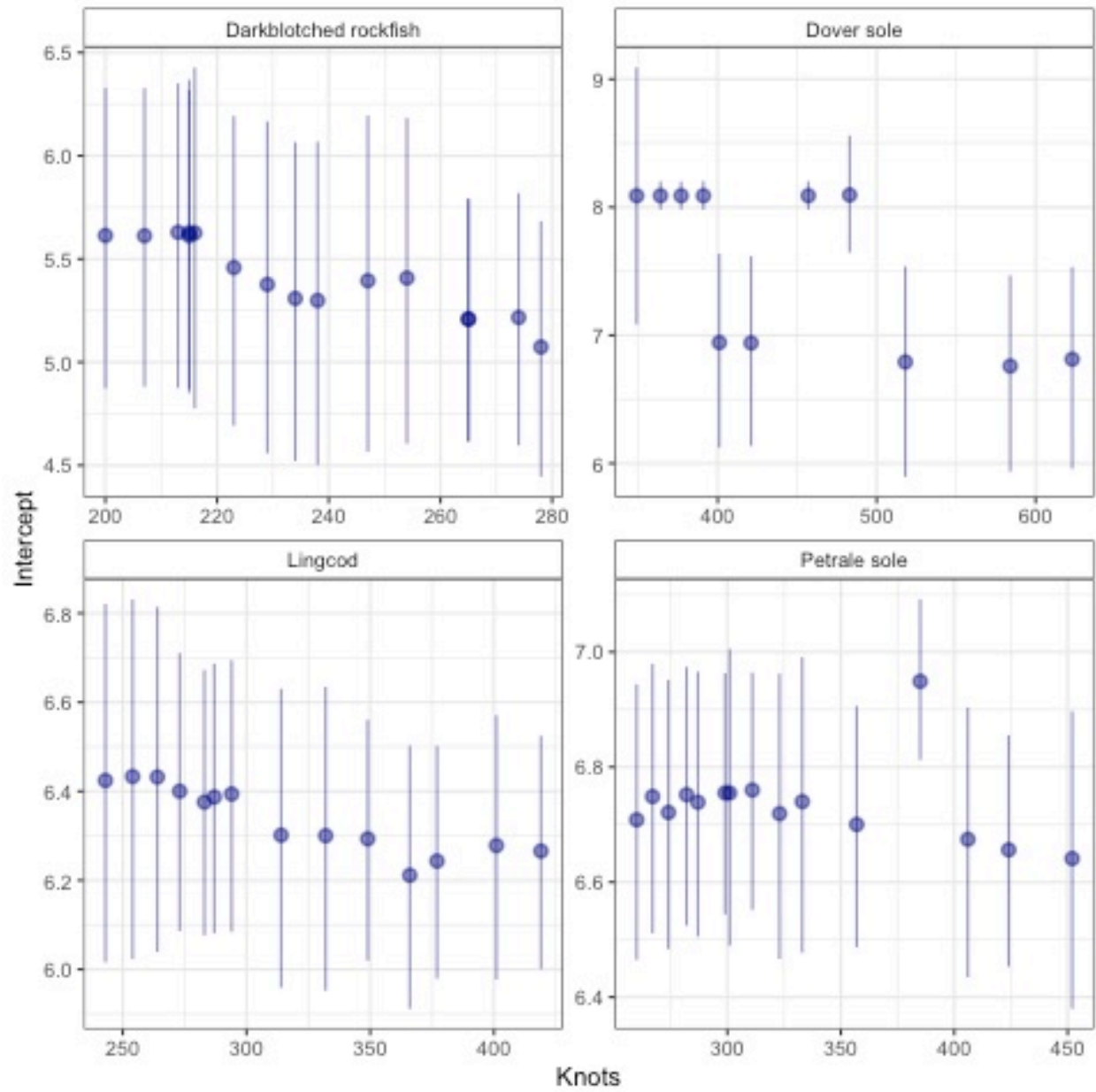


Figure 3: Figure S2

References

- Anderson, S.C. & Ward, E.J. (2019). Black swans in space: Modeling spatiotemporal processes with extremes. *Ecology*, **100**, e02403.
- Bachl, F.E., Lindgren, F., Borchers, D.L. & Illian, J.B. (2019). Inlabru: An R package for Bayesian spatial modelling from ecological survey data. *Methods in Ecology and Evolution*, **10**, 760–766.
- Bakka, H., Rue, H., Fuglstad, G.-A., Riebler, A., Bolin, D., Illian, J., Krainski, E., Simpson, D. & Lindgren, F. (2018). Spatial modeling with R-INLA: A review. *WIREs Computational Statistics*, **10**, e1443.
- Banerjee, S. & Fuentes, M. (2012). Bayesian modeling for large spatial datasets. *WIREs Computational Statistics*, **4**, 59–66.
- Datta, A., Banerjee, S., Finley, A.O. & Gelfand, A.E. (2016). Hierarchical Nearest-Neighbor Gaussian Process Models for Large Geostatistical Datasets. *Journal of the American Statistical Association*, **111**, 800–812.
- Forney, K.A., Ferguson, M.C., Becker, E.A., Fiedler, P.C., Redfern, J.V., Barlow, J., Vilchis, I.L. & Ballance, L.T. (2012). Habitat-based spatial models of cetacean density in the eastern Pacific Ocean. *Endangered Species Research*, **16**, 113–133.
- Godana, A.A., Mwalili, S.M. & Orwa, G.O. (2019). Dynamic spatiotemporal modeling of the infected rate of visceral leishmaniasis in human in an endemic area of Amhara regional state, Ethiopia. *PLOS ONE*, **14**, e0212934.
- Gómez-Rubio, V. *Bayesian inference with INLA*.
- Held, L., Schrödle, B. & Rue, H. (2010). Posterior and Cross-validatory Predictive Checks: A Comparison of MCMC and INLA. *Statistical Modelling and Regression Structures: Festschrift in Honour of Ludwig Fahrmeir* (eds T. Kneib & G. Tutz), pp. 91–110. Physica-Verlag HD, Heidelberg.
- Huang, J., Malone, B.P., Minasny, B., McBratney, A.B. & Triantafyllis, J. (2017). Evaluating a Bayesian modelling approach (INLA-SPDE) for environmental mapping. *The Science of the Total Environment*, **609**, 621–632.
- Johnson, K.F., Thorson, J.T. & Punt, A.E. (2019). Investigating the value of including depth during spatiotemporal index standardization. *Fisheries Research*, **216**, 126–137.
- Keller, A.A., Wallace, J.R. & Methot, R.D. (2017). *The Northwest Fisheries Science Center’s West Coast Groundfish Bottom Trawl Survey: History, Design, and Description*. Seattle, WA.

- Kristensen, K., Nielsen, A., Berg, C.W., Skaug, H. & Bell, B.M. (2016). TMB: Automatic Differentiation and Laplace Approximation. *Journal of Statistical Software*, **70**, 1–21.
- Latimer, A.M., Banerjee, S., Jr, H.S., Mosher, E.S. & Jr, J.A.S. (2009). Hierarchical models facilitate spatial analysis of large data sets: A case study on invasive plant species in the northeastern United States. *Ecology Letters*, **12**, 144–154.
- Lindgren, F. & Rue, H. (2015). Bayesian Spatial Modelling with R-INLA. *Journal of Statistical Software*, **63**, 1–25.
- Lindgren, F., Rue, H. & Lindström, J. (2011). An explicit link between Gaussian fields and Gaussian Markov random fields: The stochastic partial differential equation approach. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, **73**, 423–498.
- Miller, D.L., Burt, M.L., Rexstad, E.A. & Thomas, L. (2013). Spatial models for distance sampling data: Recent developments and future directions. *Methods in Ecology and Evolution*, **4**, 1001–1010.
- Murase, H., Nagashima, H., Yonezaki, S., Matsukura, R. & Kitakado, T. (2009). Application of a generalized additive model (GAM) to reveal relationships between environmental factors and distributions of pelagic fish and krill: A case study in Sendai Bay, Japan. *ICES Journal of Marine Science*, **66**, 1417–1424.
- Mutanga, O., Adam, E. & Cho, M.A. (2012). High density biomass estimation for wetland vegetation using WorldView-2 imagery and random forest regression algorithm. *International Journal of Applied Earth Observation and Geoinformation*, **18**, 399–406.
- Osgood-Zimmerman, A. & Wakefield, J. (2021). A Statistical Introduction to Template Model Builder: A Flexible Tool for Spatial Modeling.
- R Core Team. (2021). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing, Vienna, Austria.
- Righetto, A.J., Faes, C., Vandendijck, Y. & Ribeiro, P.J. (2020). On the choice of the mesh for the analysis of geostatistical data using R-INLA. *Communications in Statistics - Theory and Methods*, **49**, 203–220.
- Ross, B.E., Hooten, M.B. & Koons, D.N. (2012). An Accessible Method for Implementing Hierarchical Models with Spatio-Temporal Abundance Data. *PLOS ONE*, **7**, e49395.
- Rue, H., Martino, S. & Chopin, N. (2009). Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, **71**, 319–392.

206 Simpson, D.P., Rue, H., Martins, T.G., Riebler, A. & Sørbye, S.H. (2015). Penalising model component
207 complexity: A principled, practical approach to constructing priors. *arXiv:1403.4630 [stat]*.

208 Thorson, J.T. (2019a). Guidance for decisions using the Vector Autoregressive Spatio-Temporal (VAST)
209 package in stock, ecosystem, habitat and climate assessments. *Fisheries Research*, **210**, 143–161.

210 Thorson, J.T. (2019b). Measuring the impact of oceanographic indices on species distribution shifts: The
211 spatially varying effect of cold-pool extent in the eastern Bering Sea. *Limnology and Oceanography*, **64**,
212 2632–2645.

213 Thorson, J.T. & Barnett, L.A.K. (2017). Comparing estimates of abundance trends and distribution shifts
214 using single- and multispecies models of fishes and biogenic habitat. *ICES Journal of Marine Science*,
215 **74**, 1311–1321.

216 Tolimieri, N., Wallace, J. & Haltuch, M. (2020). Spatio-temporal patterns in juvenile habitat for 13 ground-
217 fishes in the California Current Ecosystem. *PLOS ONE*, **15**, e0237996.

218 Valavi, R., Elith, J., Lahoz-Monfort, J.J. & Guillera-Arroita, G. (2019). blockCV: An r package for generat-
219 ing spatially or environmentally separated folds for k-fold cross-validation of species distribution models.
220 *Methods in Ecology and Evolution*, **10**, 225–232.

221 Williamson, L.D., Scott, B.E., Laxton, M.R., Bachl, F.E., Illian, J.B., Brookes, K.L. & Thompson, P.M.
222 Spatiotemporal variation in harbor porpoise distribution and foraging across a landscape of fear. *Marine*
223 *Mammal Science*, **n/a**.

224 Wilson, B. (2021). Evaluating the INLA-SPDE approach for Bayesian modeling of earthquake damages from
225 geolocated cluster data.

226 Wood, S.N. (2003). Thin plate regression splines. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, **65**, 95–114.